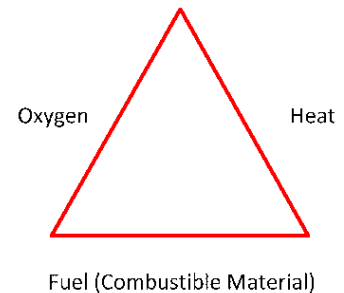

FIRE SAFETY AND EMERGENCY RESPONSE

Fire and smoke inhalation is a major cause of preventable accidental death in the workplace, yet most people ignore or are not aware of the risk that fire presents. Workplace fires occur every day and we need to know what to do if a fire takes place.

What is Fire?

Fire is a chemical reaction that produces heat, light, flame, ash, and smoke. There are 3 major elements or factors that support the start of a fire. These are: **Heat** (source of energy), **Oxygen**, and **Fuel** (combustible material like firewood, paper, grass, petrol etc).



Methods of Putting Out the Fire

They are 3 methods of extinguishing fire. These are:

- **Starvation:** this involves taking away the fuel by removal of the unburnt combustible material away from the burning material;
- **Suffocation/smothering:** this involves cutting off the supply of oxygen to the fire; and
- **Cooling:** this involves reducing the temperature. Using a lot of water on solid materials normally puts out the fire.

Classification of Fires & What to Use to Fight Them

Most fires are simply where a solid material has caught fire. However, some fires can be started by liquids and gases that can be highly flammable as well as the equipment we use every day. Fire has been classified according to the different types of matter: solids, liquids and gases that provide the fuel source.

1. **Class 'A' fires (solids):** These are fires fuelled by solid materials such as paper, wood, grass, textile etc. The way of fighting these fires is usually by *cooling*, for instance by using water extinguishers. *Suffocation* also works well therefore Dry Powder and Foam extinguishers are excellent for these types of fires as they prevent air feeding the flames.
2. **Class 'B' fires (liquids):** These are fires fuelled by highly flammable liquids like cooking oil and fats, diesel, kerosene, and petrol. The way of fighting these fires is by *suffocating* the fire, by using Dry Powder or Foam (both make an air tight barrier) or CO₂ extinguishers. The best means of extinguishing oil or cooking pan fires (such as when deep frying potatoes) is the Fire Blanket. **Water should *never* be put on class 'B' fires.** Note that you may see the classification 'F' referring to fat fires (such as cooking pans) but for our purposes we can put the two together under 'B'
3. **Class 'C' fires (gases):** These are fires fuelled by highly flammable gases such as propane, butane, acetylene, etc. If this type of fire occurs, everyone must evacuate the area very quickly. However, if it is a small flame such as an out of control cooker burner CO₂ extinguishers work very well – remember to turn off the source of the gas (to *starve* the fire) as soon as it is safe to do so!
4. **Class 'E' fires (electrical):** These are fires that caused by or involving electrical sources. These fires are ideally fought by *suffocation*: Dry Powder and CO₂ extinguishers are best for these. Dry powder will leave a sticky residue on the equipment destroying the equipment in the process but is very

effective. CO₂ extinguishers however leave no such residue and is therefore the best option for most electrical fires – if in doubt use one or the other. **Never use liquids on an electrical fire.** Remember to turn off the electric supply as soon as it is safe to do so.

How to Prevent a Fire

Prevention of something that may cause harm or damage is better than cure and fires are no exception. By adopting a proactive approach to fire safety in your workplace and home you can limit the risk of fire happening. The following are a few basic precautions everybody can take to help combat a fire risk:

- Keep storage and working areas free of rubbish. Place oily rags, waste paper etc in covered containers.
- Ensure that smoking is only allowed in designated areas.
- Do not refuel gasoline-powered equipment in a confined space, especially in the presence of an open flame such as a furnace or water heater and always turn the appliance off first.
- Do not refuel gasoline-powered equipment while it is hot.
- Keep flammable liquids stored in tightly closed (ideally self-closing) spill-proof containers.
- Pour from storage drums only what you need.
- Store flammable liquids away from spark-producing sources and living quarters.
- Use flammable liquids only in well-ventilated areas: gasses arising from a flammable liquid may themselves be explosive.
- Look for old wiring, worn insulation and broken electrical fitting, and report any hazardous condition to a manager. Do not use any electrical equipment that you believe may be unsafe.
- Prevent motors (such as fans) from overheating by keeping them clean and in good working order. A spark from a rough-running motor can ignite the oil and dust in it.
- Do not misuse fuses. Never install a fuse rated higher than specified for the circuit.
- Investigate any appliance or electrical equipment that smells strange; unusual odours can be the first sign of fire.
- Do not overload wall outlets. One outlet should have no more than two plugs.

When to Fight a Fire

If you think the fire is small enough for you to tackle you need to make a simple but quick decision to fight the fire yourself or to raise the alarm and get help. The first question you need to ask is, 'What is the fire burning... what is the fuel source?' This will determine whether you have the equipment to fight the fire.

Remember the guide about the different classes? Can you remember them? If not, stop reading now and go back to that section and re-read it: you won't have time after the fire has started!

Only fight a fire if you feel it is safe to do so!

How to Identify the Correct Fire Extinguisher

Once you have established what Class the fire is you will know what extinguisher is best suited to the task of putting the fire out. Color schemes are used on the containers of the extinguishers, but these are not globally consistent. Labelling will specify the type of extinguisher and staff must be familiar with what type of extinguisher is present and, particularly, what types of fire the extinguisher **must not** be used on (e.g., a water extinguisher must not be used on live electrical equipment).

The following is an easy follow guide to the types of equipment you will find.

Extinguisher	Best For	Danger	How To Use	How It Works
Water	Wood, cloth, paper, plastics, coal etc. Fires involving solids. Do not use on liquids such as grease, fats, oil, paint, petrol, etc.	Do not use on live electrical equipment	Point the jet or discharge horn at the base of the flames and, with a rapid sweeping motion, drive the fire towards the far edge until all the flames are out.	Has a cooling action on solid combustible materials.
Multi-Purpose dry powder	Wood, cloth, paper, plastics, coal etc. Fires involving solids. Liquids such as grease, fats, oil, paint, petrol, etc. but not on cooking fat / pan fires.	Safe on live electrical equipment, although it does not penetrate the spaces in equipment easily and the fire may re-ignite. This type of extinguisher does not cool the fire very well and care should be taken that the fire does not flare up again. Smouldering material in deep seated fires such as upholstery or bedding can cause the fire to start up again.	Point the jet or discharge horn at the base of the flames and, with a rapid sweeping motion, drive the fire towards the far edge until all the flames are out. If the extinguisher has a shut-off control wait until the air clears and if you can still see the flames then attack the fire again.	Knocks down flames and, on burning solids, melts to form a skin smothering the fire. It provides some cooling effect.
AFFF (Aqueous film-forming foam) (multi-purpose)	Wood, cloth, paper, plastics, coal etc. Fires involving solids. Liquids such as grease, fats, oil, paint, petrol, etc. but not on cooking fat / pan fires.	Do not use on cooking fat / pan fires unless you are specifically trained to do so.	For fires involving solids point the jet at the base of the flames and keep it moving across the area of the fire. Ensure that all areas of the fire are out. For fire involving liquids, do not aim the jet straight into the liquid. Where the liquid on fire is in a container, point the jet on a nearby surface above the burning liquid. Allow the foam to build up and flow across the liquid.	Forms a fire extinguishing film on the surface of a burning liquid. It has a cooling action with a wider extinguishing application than water on solid combustible materials.

Extinguisher	Best For	Danger	How To Use	How It Works
Carbon Dioxide CO ₂	Electrical fires, liquids such as grease, fats, oil paint, petrol etc.	This type of extinguisher does not cool the fire very well and you need to watch that the fire does not start up again. Fumes from CO ₂ extinguishers can be harmful if used in confined spaces: ventilate the area as soon as the fire has been controlled. The end of the horn gets very cold and can burn your hands, only hold at hand grip.	The discharge horn should be directed at the base of the flames and the jet kept moving across the area of the fire.	Vaporizing liquid gas which smothers the flames by displacing oxygen in the air.
 Fire Blanket	Fires involving both solids and liquids. Best for small fires in textiles and for cooking fat / pan fires.	If the blanket does not completely cover the fire, it will not extinguish the fire.	Place carefully over the fire. Keep your hands shielded from the fire. Do not waft the fire towards you.	Smothers the flame, removing oxygen from the fuel and heat.

How to use the Fire Extinguisher

Remember the acronym: "**P.A.S.S**"

P... Pull the pin.

A... Aim the extinguisher nozzle at the base of the flames.

S... Squeeze trigger while holding the extinguisher upright.

S... Sweep the extinguisher from side to side, covering the area of the fire with the extinguishing agent.

When NOT to Fight a Fire

- When the fire is spreading quickly beyond the spot where it started.
- When you are not sure of the location of your escape exit – use the time to find a way out.
- If you do not have adequate fire-fighting equipment or experience.
- If your path of escape is threatened, if the extinguisher runs out, if the extinguisher proves to be ineffective, or if you will no longer be able to safely fight the fire for whatever reason...**LEAVE THE AREA IMMEDIATELY!**

The Immediate Actions

- Remain calm.
- Raise the alarm advising those around to evacuate and alert others shouting “FIRE! FIRE! FIRE!”
- If it is safe to do so then secure vital records and shut down electrical equipment.
- Proceed to your pre-determined exit if necessary. Shut all doors as you leave – do not lock them.
- Ensure that all staff and visitors evacuate the area.
- Assist disabled staff and/or visitors.
- Proceed in an orderly manner to the assembly point and conduct a head count.
- DO NOT OPEN YOUR DOOR if it is hot or if smoke is present.
- Test to see if your door is hot using the back of your hand.

DO NOT PANIC! STAY CALM AND ACT DECISIVELY.

- The last one out of the room should not lock the door, just close it. Locking the door hinders the fire search and rescue efforts.
- Stay low to avoid smoke and toxic gases. The best air is close to the floor, so crawl if necessary.
- If possible, cover your mouth and nose with a damp cloth to help you in breathing.

What to do if you are Trapped in a Burning Building

- Alert people immediately.
- Do not panic.
- If possible, control the fire with an appropriate extinguisher if training, experience, or common sense tells you that it is safe to do so.
- Do not take any personal risk. Fight the fire from a position where you can escape. Take no risk if the fumes from the burning material are unknown to you.
- Check your room door. If it is hot do not open it.
- If it is not hot then you should open the door slowly and be prepared to close it at once if the corridor is full of smoke or fire.
- If there is no smoke or fire in the hallway then move quickly to the exit stairways or door and out of the building.
- If there is smoke in the hallway then crawl on your hands and knees to the exit.
- If the exit is smoke-filled then go to a safe exit.
- If you cannot make it to the exit then keep your door closed and place cloth material (ideally dampened with water) around and under the door to prevent smoke from entering.
- Close as many doors as possible between you and the fire.
- DO NOT open or break windows unless necessary to escape as the open window may act as a chimney. If breathing is difficult then you can try to ventilate the room to allow poisonous fumes to escape by opening the top portion of the window or partially opening the main window.

Further instructions